

$$C_2 = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 5 \\ -7 \end{bmatrix} = \begin{bmatrix} -7 \cdot 5 + (-5) \cdot (-7) \\ 3 \cdot 5 + 2 \cdot (-7) \end{bmatrix} \quad (\text{Row-Vector Rule for Computing } Ax) \\ = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

$$\therefore CA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{Thus } C = A^{-1}. \quad \square$$

Theorem 4: Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. If $ad - bc \neq 0$, then A is invertible and $A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. If $ad - bc = 0$, then A is not invertible.

The proof of Theorem 4 must be done in Exercises 25 and 26.

determinant of A is denoted $\det A$. $\boxed{\det A = ad - bc}$.

Theorem 4 says that a 2×2 matrix A is invertible if and only if $\det A \neq 0$.

Example 2: Find the inverse of $A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$.

Solution: $\det A = 3 \cdot 6 - 5 \cdot 4 = -2 \neq 0$. By Theorem 4, A is invertible and $A^{-1} = -\frac{1}{2} \begin{bmatrix} 6 & -4 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix} \quad \square$

Theorem 5: If A is an invertible $n \times n$ matrix, then for each b in $\mathbb{R}^n$, the equation $Ax = b$ has the unique solution $x = A^{-1}b$.

Proof: By definition of inverse, there exists a ^{unique} $n \times n$ matrix A^{-1} s.t. $A^{-1}A = AA^{-1} = I_n$.

Let, $Ax = b$, where b is an arbitrary vector in $\mathbb{R}^n$ (Theorem 2(a))

$$A^{-1}(Ax) = A^{-1}b \Rightarrow (A^{-1}A)x = A^{-1}b \Rightarrow I_n x = A^{-1}b \quad (\text{By defn. of identity matrix}) \\ \Rightarrow \boxed{x = A^{-1}b}$$

~~$A^{-1}Ax = A^{-1}b$ is a solution of the equation $Ax = b$.~~